

Physics All-Topics Review — Vectors, Kinematics and Forces

One note for everything so far. Built from Sessions S1–S6: vectors, motion in a line, motion graphs, projectiles, relative velocity, circular motion, Newton’s three laws, free-body diagrams, friction, inclines and connected objects.

Conventions: $g = 9.81 \text{ m/s}^2$ · up and right are positive · u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement · angles measured from the $+x$ axis unless stated.

How to use it (45 minutes): 10 min section 1 (vectors) · 10 min sections 2–3 · 10 min sections 4–5 · 10 min sections 6–7 · 5 min section 8 (mixed check).

1. Vectors

1.1 Scalar or vector

	Scalar	Vector
Has	size (magnitude) only	size and direction
Examples	distance, speed, time, mass, energy, temperature	displacement, velocity, acceleration, force, weight
Adds by	ordinary arithmetic	tip-to-tail or components

A vector is drawn as an arrow: length = magnitude (to scale), arrowhead = direction. A **negative** vector has the same length and points the opposite way.

1.2 Adding vectors

- **Same line:** add with signs. $+8 \text{ N}$ and -3 N give $+5 \text{ N}$.
- **Tip-to-tail (drawing):** put the tail of the second vector at the tip of the first. The **resultant** runs from the first tail to the last tip.
- **Perpendicular vectors** (Pythagoras):

$$R = \sqrt{A^2 + B^2} \quad \tan \theta = \frac{B}{A}$$

Why: tip-to-tail, two perpendicular vectors form the legs of a right triangle and the resultant is the hypotenuse.

1.3 Components

Any vector A at angle θ from the x -axis is the sum of a horizontal and a vertical part:

$$A_x = A \cos \theta \quad A_y = A \sin \theta$$

Why: in the right triangle with hypotenuse A , $\cos \theta = \frac{A_x}{A}$ and $\sin \theta = \frac{A_y}{A}$.

Back from components to the vector:

$$A = \sqrt{A_x^2 + A_y^2} \quad \theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

Signs by quadrant: left means $A_x < 0$, down means $A_y < 0$. Always check the angle against a sketch (the calculator's $\tan^{-1}$ only returns angles between -90° and 90°).

1.4 The component method (any two or more vectors)

1. Resolve every vector into x and y components.
2. Add: $R_x = A_x + B_x$, $R_y = A_y + B_y$.
3. Rebuild: $R = \sqrt{R_x^2 + R_y^2}$, $\theta = \tan^{-1} (R_y/R_x)$.

Example. $A = 5.0 \text{ m at } 30^\circ$, $B = 3.0 \text{ m at } 120^\circ$.

	x	y
A	$5.0 \cos 30^\circ = 4.33$	$5.0 \sin 30^\circ = 2.50$
B	$3.0 \cos 120^\circ = -1.50$	$3.0 \sin 120^\circ = 2.60$
R	2.83	5.10

$$R = \sqrt{2.83^2 + 5.10^2} = 5.83 \text{ m}, \theta = \tan^{-1} (5.10/2.83) = 61^\circ \text{ above the } +x \text{ axis.}$$

Subtracting: $A - B = A + (-B)$. Reverse B , then add.

Equilibrium: if the vectors add to zero, the object is in equilibrium ($R_x = 0$ and $R_y = 0$).

1.5 Scale drawings

Choose a scale (for example 1 cm : 10 N), draw each vector with ruler and protractor, measure the resultant, and convert back. Real size = drawing size $\times$ scale factor.

2. Motion in a straight line

Quantity	Definition	Type
Distance	total path length	scalar
Displacement s	change in position, start to finish	vector
Average speed	$\frac{\text{distance}}{\text{time}}$	scalar
Average velocity	$v_{avg} = \frac{\Delta x}{\Delta t}$	vector
Acceleration	$a = \frac{\Delta v}{\Delta t} = \frac{v - u}{t}$	vector

Speeding up: a and v have the same sign. Slowing down: opposite signs. Negative acceleration does not always mean slowing down.

2.1 The kinematic equations (constant acceleration only)

S1. From the definition of acceleration, $a = \frac{v - u}{t}$:

$$v = u + at$$

S0. With constant a , velocity changes evenly, so the average velocity is $\frac{u + v}{2}$:

$$s = \frac{1}{2} (u + v) t$$

S2. Put $v = u + at$ into S0:

$$s = \frac{1}{2} (u + u + at) t = ut + \frac{1}{2}at^2$$

S3. From S1, $t = \frac{v-u}{a}$. Put it into S0: $s = \frac{(v+u)(v-u)}{2a} = \frac{v^2-u^2}{2a}$, so

$$v^2 = u^2 + 2as$$

Choosing: list the five quantities s, u, v, a, t . You know three and want one; use the equation that does not contain the fifth.

Example. A car goes from rest to 25 m/s in 10 s. $a = \frac{25-0}{10} = 2.5 \text{ m/s}^2$;
 $s = \frac{1}{2} (0 + 25) (10) = 125 \text{ m}$.

2.2 Free fall

Only gravity acts: $a = -g = -9.81 \text{ m/s}^2$ (up positive), the same for every mass.

- At the top of the flight $v = 0$, but a is still -9.81 m/s^2 .
- Time up = time down, and the speed is the same at the same height.

Example. Dropped from 45 m: $s = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2(45)}{9.81}} = 3.03 \text{ s}$; $v = gt = 29.7 \text{ m/s}$.

2.3 Motion graphs

Graph	Slope gives	Area under the line gives
position-time	velocity	—
velocity-time	acceleration	displacement
acceleration-time	—	change in velocity

Position-time: straight line = constant velocity, curve = accelerating, flat = at rest. Velocity-time: flat = constant velocity, straight sloped line = constant acceleration, crossing the axis = changing direction.

3. Projectile motion

Key idea: horizontal and vertical motions are independent. Horizontal: constant velocity ($a_x = 0$). Vertical: free fall ($a_y = -g$). They share the same time t .

Launch speed u at angle θ : $u_x = u \cos \theta$, $u_y = u \sin \theta$.

$$x = u_x t \quad v_y = u_y - gt \quad y = u_y t - \frac{1}{2}gt^2$$

3.1 Launched from the ground, landing at the same height

- **Time of flight.** Set $y = 0$: $t (u_y - \frac{1}{2}gt) = 0$, so $T = \frac{2u \sin \theta}{g}$.
- **Maximum height.** At the top $v_y = 0$; from $v_y^2 = u_y^2 - 2gH$: $H = \frac{u^2 \sin^2 \theta}{2g}$.
- **Range.** $R = u_x T = \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin 2\theta}{g}$. Largest at 45° ; angles that add to 90° give the same range.

Example. $u = 20 \text{ m/s}$ at 30° : $u_x = 17.3$, $u_y = 10.0 \text{ m/s}$. $T = \frac{2(10.0)}{9.81} = 2.04 \text{ s}$;
 $H = \frac{10.0^2}{2(9.81)} = 5.10 \text{ m}$; $R = 17.3 \times 2.04 = 35.3 \text{ m}$.

3.2 Launched horizontally from a height h

$u_y = 0$, so the fall time depends only on the height:

$$t = \sqrt{\frac{2h}{g}} \quad x = u_x t \quad v_y = gt \quad \text{speed} = \sqrt{u_x^2 + v_y^2}$$

Example. 15 m/s off a 20 m cliff: $t = \sqrt{\frac{2(20)}{9.81}} = 2.02 \text{ s}$; $x = 15 \times 2.02 = 30.3 \text{ m}$; $v_y = 19.8 \text{ m/s}$;
 impact speed $= \sqrt{15^2 + 19.8^2} = 24.8 \text{ m/s}$.

3.3 Launched at an angle from a height

Take the launch point as the origin. Landing at depth h below it means $y = -h$:

$$-h = u_y t - \frac{1}{2}gt^2$$

Solve this quadratic for t (keep the positive root), then $x = u_x t$.

4. Relative velocity

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

Read it as “velocity of A relative to C = A relative to B + B relative to C”. The inner letters match. Also
 $\vec{v}_{AB} = -\vec{v}_{BA}$.

- **Same line:** two cars at 30 m/s and 20 m/s . Same direction: relative speed 10 m/s . Opposite directions: 50 m/s .
- **Boat crossing a river:** boat 4.0 m/s straight across, current 3.0 m/s . Speed relative to the bank $= \sqrt{4.0^2 + 3.0^2} = 5.0 \text{ m/s}$, at $\tan^{-1}(3.0/4.0) = 36.9^\circ$ downstream of straight across. River 100 m wide: crossing time $= \frac{100}{4.0} = 25 \text{ s}$ (uses only the across component); drift downstream $= 3.0 \times 25 = 75 \text{ m}$.

5. Uniform circular motion

Constant speed, but the direction keeps changing, so the object **is accelerating**, toward the center.

$$v = \frac{2\pi r}{T} \quad a_c = \frac{v^2}{r} \quad F_c = \frac{mv^2}{r}$$

- T = period (time for one lap); frequency $f = \frac{1}{T}$.
- Velocity is tangent to the circle. Acceleration and net force point to the center (**centripetal**).
- Centripetal force is not a new force. It is the net force supplied by tension, friction, gravity or the normal force. If it vanishes, the object moves off in a straight line along the tangent.

Example. $r = 0.50 \text{ m}$, $T = 2.0 \text{ s}$, $m = 0.20 \text{ kg}$: $v = \frac{2\pi(0.50)}{2.0} = 1.57 \text{ m/s}$; $a_c = \frac{1.57^2}{0.50} = 4.93 \text{ m/s}^2$;
 $F_c = 0.20 \times 4.93 = 0.99 \text{ N}$.

6. Newton's laws of motion

First law (inertia). An object at rest stays at rest, and an object in motion keeps moving at constant velocity, unless a net external force acts on it.

$$\sum \vec{F} = 0 \Leftrightarrow \vec{v} = \text{constant} \quad (\vec{a} = 0)$$

Inertia is the resistance to a change in motion; mass measures it. Constant velocity means balanced forces, not “no forces”.

Second law. The acceleration of an object is directly proportional to the net force on it, inversely proportional to its mass, and in the direction of the net force.

$$\vec{F}_{net} = m\vec{a} \quad 1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

Apply it separately in each direction: $\sum F_x = ma_x$, $\sum F_y = ma_y$.

Third law. When object A exerts a force on object B, B exerts a force on A that is equal in size and opposite in direction.

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

The two forces act on **different objects**, so they never cancel each other. They are the same type of force and exist together. (A book's weight and the normal force from the table are *not* a third-law pair: both act on the book.)

Mass and weight. Mass (kg) is the amount of matter and is the same everywhere. Weight is the force of gravity: $W = mg$ (N).

Apparent weight (elevator). The scale reads the normal force. From $N - mg = ma$:

$$N = m(g + a)$$

Accelerating up: heavier. Accelerating down: lighter. Constant velocity: $N = mg$. Free fall ($a = -g$): $N = 0$.

Terminal velocity. Air drag grows with speed. When drag equals weight, the net force is zero and the speed stays constant.

Example. A 60 kg person in an elevator accelerating upward at 2.0 m/s^2 :
 $N = 60(9.81 + 2.0) = 709 \text{ N}$.

7. Free-body diagrams, friction, inclines, connected objects

7.1 Free-body diagram (FBD)

Draw the object as a dot. Draw only the forces acting **on** it, each as an arrow from the dot, labeled: weight mg (down), normal N (perpendicular to the surface), tension T (along the rope, away from the object), friction f (along the surface, opposing sliding), applied force. Never draw “ ma ” or velocity as a force.

Then: choose axes $\rightarrow$ resolve tilted forces into components $\rightarrow$ write $\sum F = ma$ for each axis $\rightarrow$ solve.

7.2 Friction

$$f_s \leq \mu_s N \quad f_k = \mu_k N$$

- **Static** friction matches the push exactly, up to a maximum $\mu_s N$. **Kinetic** friction is constant once sliding; usually $\mu_k < \mu_s$.
- On a level surface with only horizontal pushes, $N = mg$. If a force has a vertical part, N changes.

Example. A 10 kg box is pulled horizontally with 50 N; $\mu_k = 0.30$. $N = 10 (9.81) = 98.1$ N;
 $f_k = 0.30 (98.1) = 29.4$ N; $a = \frac{50 - 29.4}{10} = 2.06$ m/s².

7.3 Inclined plane (angle θ)

Tilt the axes along and perpendicular to the slope. Weight resolves into $mg \sin \theta$ down the slope and $mg \cos \theta$ into the slope.

Perpendicular: $N = mg \cos \theta$. Along the slope (sliding down): $mg \sin \theta - \mu_k mg \cos \theta = ma$, so

$$a = g (\sin \theta - \mu_k \cos \theta) \quad \text{frictionless: } a = g \sin \theta$$

A block just stays at rest when $mg \sin \theta = \mu_s mg \cos \theta$, that is $\tan \theta = \mu_s$.

Example. $\theta = 30^\circ$, $\mu_k = 0.20$: $a = 9.81 (0.500 - 0.20 \times 0.866) = 3.21$ m/s².

7.4 Connected objects

Same rope: same tension and the same size of acceleration. Write $F_{net} = ma$ for each object, then add the equations so T cancels.

Atwood machine ($m_2 > m_1$): $m_2 g - T = m_2 a$ and $T - m_1 g = m_1 a$. Adding:

$$a = \frac{(m_2 - m_1) g}{m_1 + m_2} \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

Block on a table (m_1) pulled by a hanging mass (m_2): $m_2 g - T = m_2 a$ and $T - \mu_k m_1 g = m_1 a$, so

$$a = \frac{(m_2 - \mu_k m_1) g}{m_1 + m_2}$$

Example. Atwood, $m_1 = 2.0$ kg, $m_2 = 3.0$ kg: $a = \frac{(1.0)(9.81)}{5.0} = 1.96$ m/s²;

$$T = \frac{2(2.0)(3.0)(9.81)}{5.0} = 23.5 \text{ N.}$$

8. Mixed check (answers below)

1. Find the components of a 50 N force at 37° above the $+x$ axis.
2. A hiker walks 6.0 km east, then 8.0 km north. Find the distance walked and the displacement.
3. A ball is thrown straight up at 14.7 m/s. Find the time to the top and the maximum height.
4. A marble rolls off a 1.25 m high table at 4.0 m/s. How far from the table does it land?
5. A plane flies 200 m/s north in a 50 m/s wind blowing east. Find its velocity relative to the ground.
6. A 1000 kg car takes a 50 m radius curve at 20 m/s. Find the centripetal acceleration and the force. What supplies it?
7. A 5.0 kg block is pushed with 20 N on a level floor. Find a with no friction, then with $\mu_k = 0.20$.
8. A 2.0 kg block slides down a frictionless 25° incline. Find its acceleration.

9. On a velocity-time graph, what do the slope and the area represent?
10. A book rests on a table. Name the third-law partner of the book's weight.

Answers, step by step

- $F_x = 50 \cos 37^\circ = 39.9 \text{ N}; F_y = 50 \sin 37^\circ = 30.1 \text{ N}.$
- Distance = $6.0 + 8.0 = 14.0 \text{ km}$. Displacement = $\sqrt{6.0^2 + 8.0^2} = 10.0 \text{ km}$ at $\tan^{-1} (8.0/6.0) = 53.1^\circ$ north of east.
- $v = u - gt = 0 \Rightarrow t = \frac{14.7}{9.81} = 1.50 \text{ s}$. $H = \frac{u^2}{2g} = \frac{14.7^2}{2(9.81)} = 11.0 \text{ m}.$
- $t = \sqrt{\frac{2(1.25)}{9.81}} = 0.505 \text{ s}; x = 4.0 \times 0.505 = 2.02 \text{ m}.$
- Speed = $\sqrt{200^2 + 50^2} = 206 \text{ m/s}$; direction $\tan^{-1} (50/200) = 14.0^\circ$ east of north.
- $a_c = \frac{20^2}{50} = 8.0 \text{ m/s}^2$; $F_c = 1000 \times 8.0 = 8000 \text{ N}$, supplied by static friction between the tires and the road.
- No friction: $a = \frac{20}{5.0} = 4.0 \text{ m/s}^2$. With friction: $f_k = 0.20 (5.0) (9.81) = 9.81 \text{ N}$;
 $a = \frac{20 - 9.81}{5.0} = 2.04 \text{ m/s}^2$.
- $a = g \sin \theta = 9.81 \sin 25^\circ = 4.15 \text{ m/s}^2$ (the mass cancels).
- Slope = acceleration. Area between the line and the time axis = displacement.
- The weight is Earth pulling the book down, so the partner is the book pulling Earth up with an equal force. (Not the normal force.)

Notes for next session

- Weakest areas to check first: signs of components in quadrants II-IV, and third-law pairs versus balanced forces.
- Not yet covered: work, energy, momentum.